

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture 3a: Introduction: Random variables

Prof. Oke

UMassAmherst

College of Engineering

September 29, 2026

Overview of Module 3

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- Lecture 3a: Introduction: Random Variables

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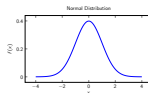
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- Lecture 3a: Introduction: Random Variables
- Lecture 3b: **Normal Distribution**

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- Lecture 3c: **Lognormal and Exponential Distributions**

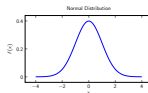


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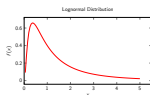
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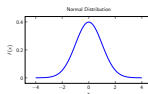


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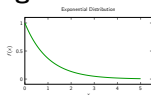
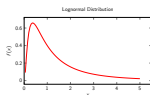
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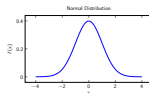
- Lecture 3d: **Binomial Distribution**

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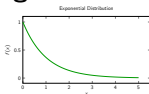
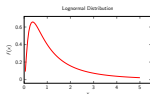
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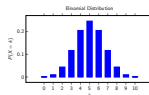
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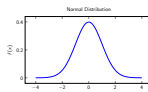
- Lecture 3e: **Poisson Distribution**

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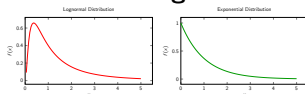
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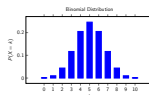
- Lecture 3b: **Normal Distribution**



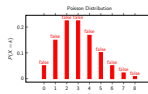
- Lecture 3c: **Lognormal and Exponential Distributions**



- Lecture 3d: **Binomial Distribution**



- Lecture 3e: **Poisson Distribution**



- Lecture 3f: Joint Distributions and further topics

Objectives and outline of today's lecture

- 1 Introduction to random variables
- 2 Probability distribution of r.v.
- 3 Discrete r.v.'s
- 4 Continuous r.v.'s
- 5 Outlook
- 6 Appendix

Objectives and outline of today's lecture

- Understand random variables

① Introduction to random variables

② Probability distribution of r.v.

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- Distinguish between discrete and continuous random variables

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- Distinguish between discrete and continuous random variables
- Compute measures of centrality and dispersion, as well as sketch PMFs, PDFs and CDFs

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- *Discrete*
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Measures of dispersion

- Variance

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Measures of dispersion

- Variance
- Standard deviation
- Coefficient of variation (COV)

Probability distribution

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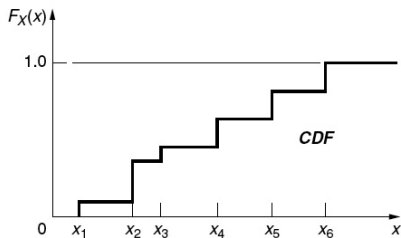
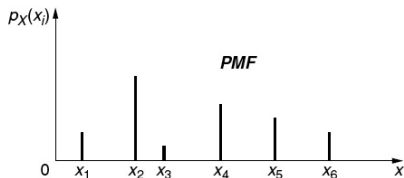
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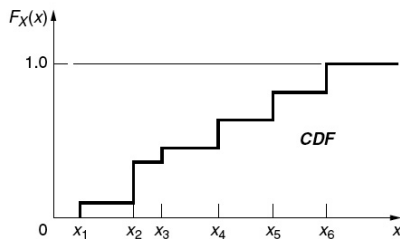
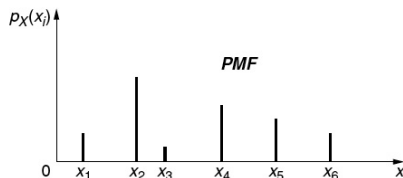
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The probability masses in a PMF sum up to 1.

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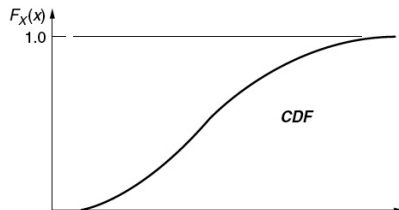
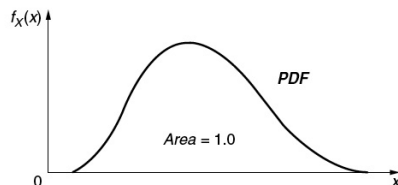
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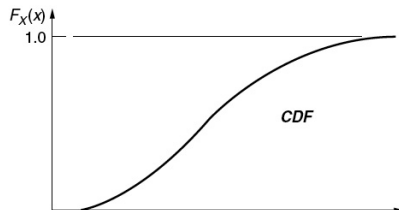
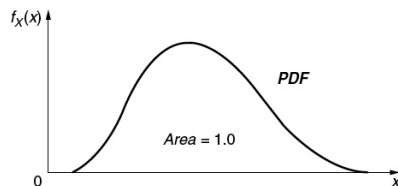
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The total area under a PDF is 1.

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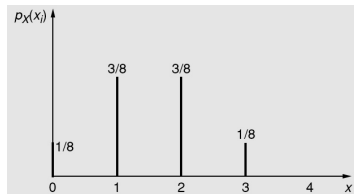


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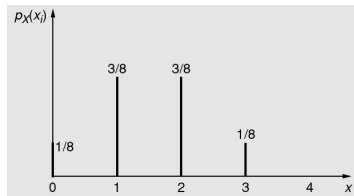


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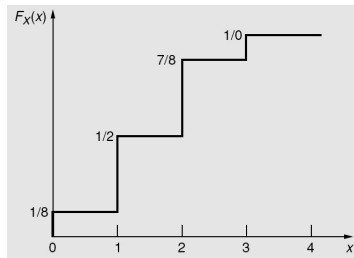


Figure: CDF

Further derivations

① Continuous case:

$$P(a < X \leq b) = \int_{-\infty}^b f_X(x) dx - \int_{-\infty}^a f_X(x) dx \quad (6)$$

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Mean and variance

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Mean

Weighted average or expected value

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Measures of dispersion (cont.)

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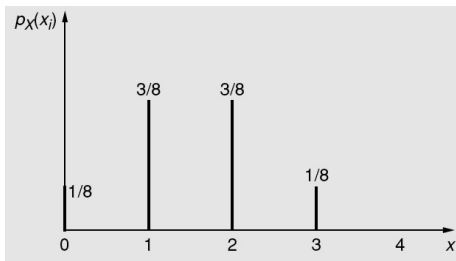
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You are given the PMF (probability mass function) of the operating condition X of bulldozers after 6 months.

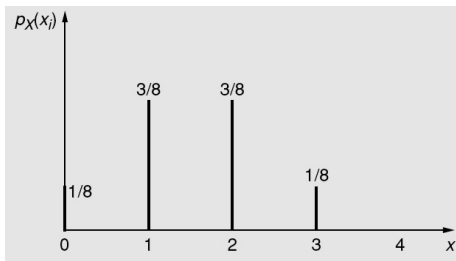
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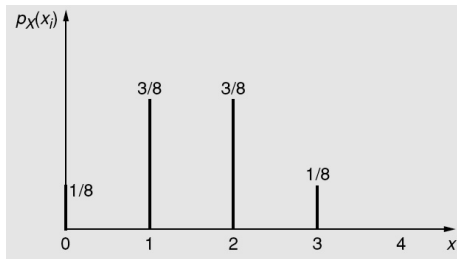
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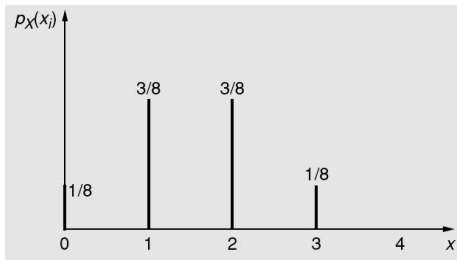
Find the mean, variance, standard deviation and coefficient of variation of X .

Example 2: Bulldozers revisited (cont.)

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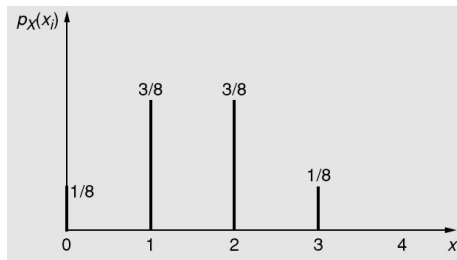


Example 2: Bulldozers revisited (cont.)



(a) Mean: $\mu_X = \mathbb{E}(X) = 0 \left(\frac{1}{8}\right) + 1 \left(\frac{3}{8}\right) + 2 \left(\frac{3}{8}\right) + 3 \left(\frac{1}{8}\right) = 1.5$.

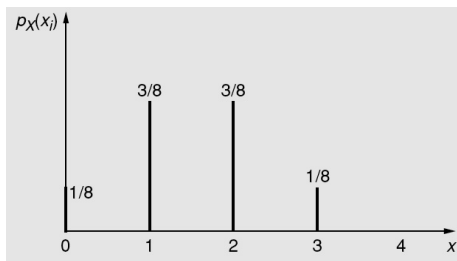
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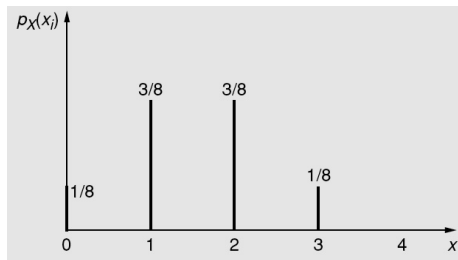
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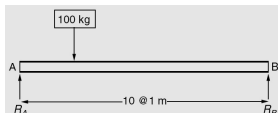


Figure E2.5a Beam AB.

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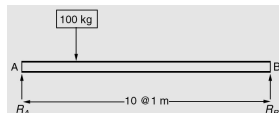


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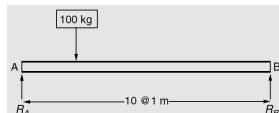


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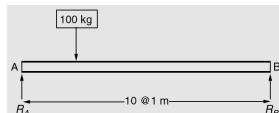


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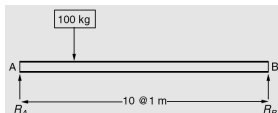


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- (a) Plot the PDF of X .
- (b) Solve the integral for the CDF and plot.
- (c) Find $P(2 < X \leq 5)$.

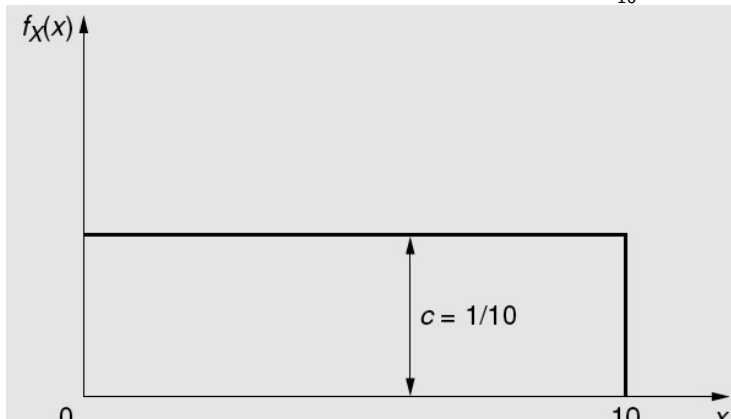
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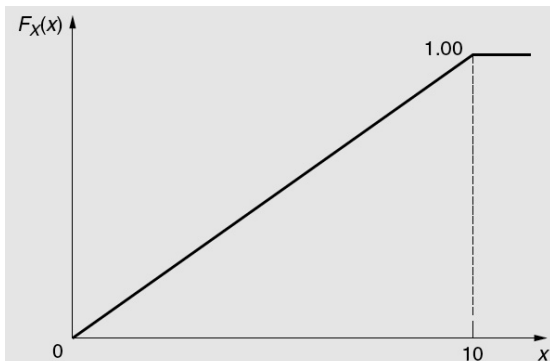


Figure E3.2b CDF of X .

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Recap

- Random variables

Reading

- Open Intro Statistics Section 3.4 (Random variables)
- Open Intro Statistics Section 3.5 (Continuous distributions)

Recap

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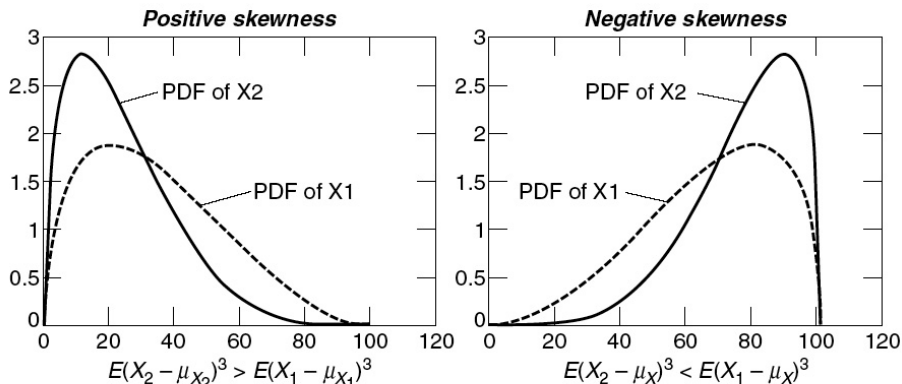
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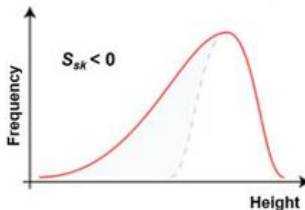
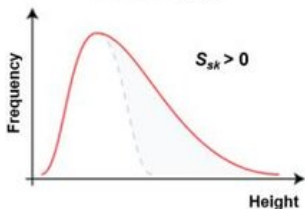
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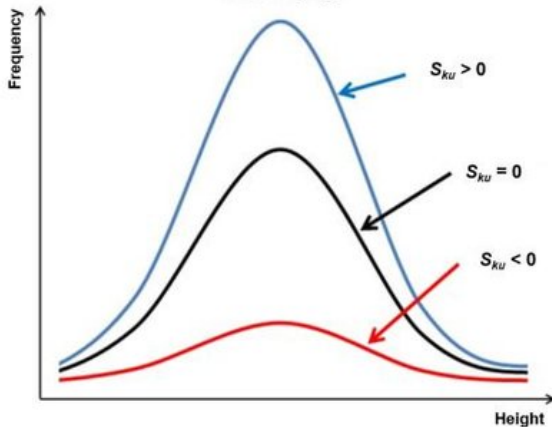
Skewness vs. kurtosis

Skewness vs. kurtosis

Skewness (S_{sk})



Kurtosis (S_{ku})



Source: Bonyar, A (2015) "Application of localization factor for the detection of tin oxidation with AFM" DOI: 10.1109/SIITME.2015.7342289

Generalized expectation

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